

ADC + DAC: Md. Jahin Alam

Schmitt trigger: Md Obaidur Rahman

Signal Generator: Quazi Shafayat Hossain

MOS+CMOS: Jishnu Mahmud

- While posting your question here, please follow this format:

|     |    |          |         |
|-----|----|----------|---------|
| (a) | CO | Question | [Marks] |
| (b) | CO | Question | [Marks] |
| (c) | CO | Question | [Marks] |

## ADC+DAC [MJA]

The figure below indicates the operation of a non-uniform Flash ADC (Analog-to-Digital Converter). A standard binary-weighted DAC (Digital-to-Analog Converter) has been connected to the outputs of the ADC.

Set : A

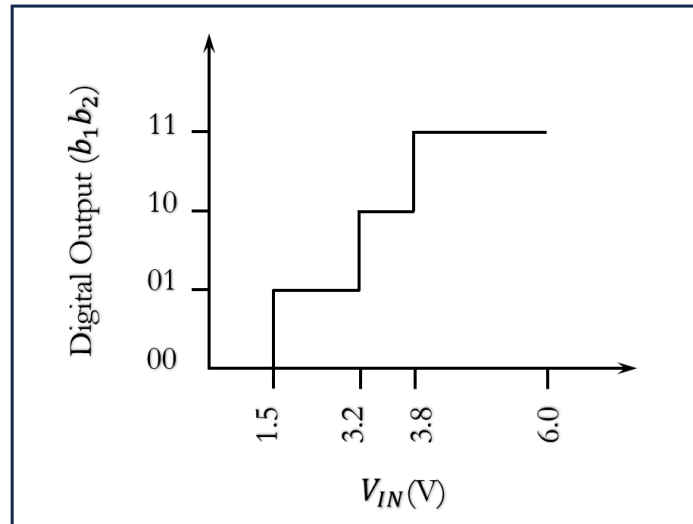


Fig : Transfer curve of a Flash ADC

Set : B

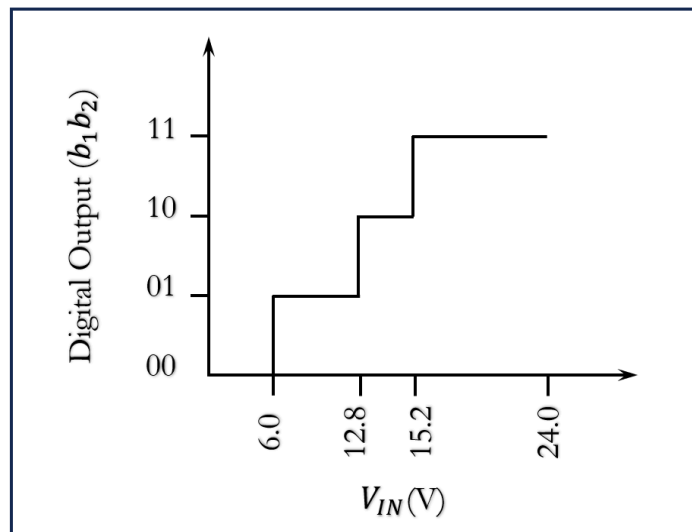


Fig : Transfer curve of a Flash ADC

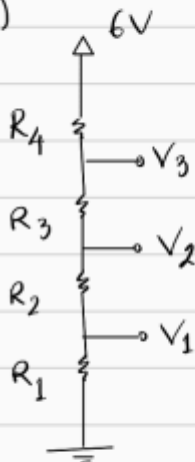
|     |     |                                                                                                                               |       |
|-----|-----|-------------------------------------------------------------------------------------------------------------------------------|-------|
| (a) | CO3 | Determine what should be the reference voltage of the ADC.                                                                    | [2]   |
| (b) | CO3 | Design, draw and label all the part values of the Flash ADC which follows the transfer curve shown above.                     | [12]  |
| (c) | CO3 | Draw the DAC mentioned in the description with the same reference voltage found in (a). Find the analog outputs from the DAC. | [2+4] |

## Solution

### Set - A

(a)  $V_{REF} = 6V$

(b)



Assume,

$$R_1 + R_2 + R_3 + R_4 = \Sigma R$$

$$\textcircled{\#} I = \frac{6}{\Sigma R}$$

$$\textcircled{\#} V_1 = 1.5 = IR_1$$

$$\Rightarrow R_1 = 0.25 \Sigma R$$

$$\textcircled{\#} V_2 - V_1$$

$$= 1.7 = IR_2$$

$$\Rightarrow R_2 = 0.28 \Sigma R$$

$$\textcircled{\#} V_3 - V_2$$

$$= 0.6 = IR_3$$

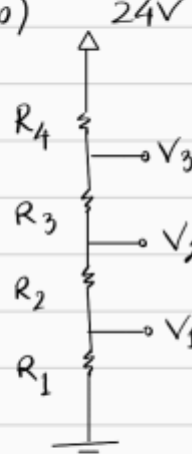
$$\textcircled{\#} R_4 = 0.37 \Sigma R$$

$$\Rightarrow R_3 = 0.12 \Sigma R$$

### Set - B

(a)  $V_{REF} = 24V$

(b)



Assume,

$$R_1 + R_2 + R_3 + R_4 = \Sigma R$$

(Same Result as Set - A) %

\* Similar type calculation as set - A :-

$$\textcircled{\#} R_1 = 0.25 \Sigma R$$

$$\textcircled{\#} R_2 = 0.28 \Sigma R$$

$$\textcircled{\#} R_3 = 0.12 \Sigma R$$

$$\textcircled{\#} R_4 = 0.37 \Sigma R$$

Draw the entire ADC  $\rightarrow \textcircled{3} \begin{cases} 3 \text{ ops} \\ 4 \text{ res} \end{cases}$

Correct Resistors  $\rightarrow \textcircled{7}$

Other Labels  $\rightarrow \textcircled{2}$

Students can take any value for  $\Sigma R = 10k, 60k, 100k$  etc.

Draw the entire ADC  $\rightarrow \textcircled{3} \begin{cases} 3 \text{ ops} \\ 4 \text{ res} \end{cases}$

Correct Resistors  $\rightarrow \textcircled{7}$

Other Labels  $\rightarrow \textcircled{2}$

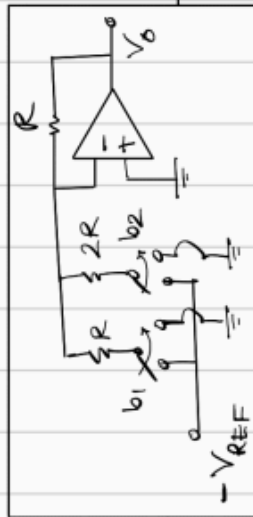
Students can take any value for  $\Sigma R = 10k, 60k, 100k$  etc.

(C) Draw Binary W. DAC → (2)

$$V_{REF} = \pm 6V$$

$$* V_o = -V_{REF} \left( b_1 + \frac{b_2}{2} \right)$$

| $b_1 b_2$ | $V_o$    |
|-----------|----------|
| 00        | $\pm 0V$ |
| 01        | $\pm 3V$ |
| 10        | $\pm 6V$ |
| 11        | $\pm 9V$ |



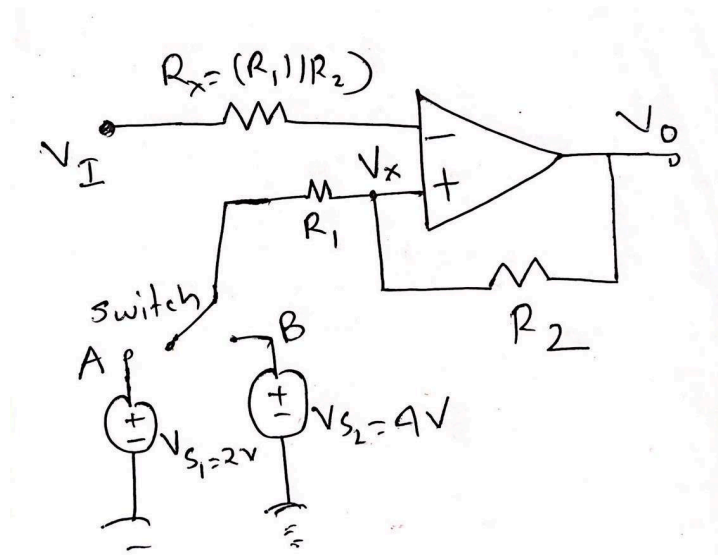
(C) Draw Binary W. DAC → (2)

$$V_{REF} = \pm 24V$$

$$* V_o = -V_{REF} \left( b_1 + \frac{b_2}{2} \right)$$

| $b_1 b_2$ | $V_o$     |
|-----------|-----------|
| 00        | $\pm 0V$  |
| 01        | $\pm 12V$ |
| 10        | $\pm 24V$ |
| 11        | $\pm 36V$ |

## Schmitt Trigger:



For the circuit shown above, the hysteresis width is 5V and the value of  $R_x$  is  $2.4\text{k}\Omega$ . Consider the following two cases:

Case 1: When the switch is in position A and the value of the input voltage  $V_I$  is **increasing initially from a very low negative value**, the voltage at  $V_x$  is 5.2V.

Case 2: When the switch is in position B and the value of the input voltage  $V_I$  is **decreasing initially from a very high positive value**, the voltage at  $V_x$  is 1.4V.

|     |     |                                                                                                                    |      |
|-----|-----|--------------------------------------------------------------------------------------------------------------------|------|
| (a) | CO1 | Find the value of $R_1$ and $R_2$ using the information given above only .                                         | [12] |
| (b) | CO1 | How many different voltages can appear at $V_o$ ? Find the values of those voltages.                               | [4]  |
| (c) | CO1 | Draw the input vs output characteristics for switch position at A. [ Clearly indicate all the values in the graph] | [4]  |

[ For faculties: in Question (a), If the students can write the equations properly- 7 marks  
 If they can find the relation between  $R_1$  and  $R_2$  with the equation - 10 marks  
 If they can find  $R_1$  and  $R_2$  -12 marks ]

Solution:

(a) Hysteresis width = 5V  $\Rightarrow V_{TH} - V_{TL} = 5$

$$\Rightarrow V_H \left( \frac{R_1}{R_1 + R_2} \right) - V_L \left( \frac{R_1}{R_1 + R_2} \right) = 5 \quad \text{--- (i)}$$

from Case 1:  $V_x = V_{ref} \left( \frac{R_2}{R_1 + R_2} \right) + V_o \left( \frac{R_1}{R_1 + R_2} \right)$

$$\Rightarrow 5.2 = 2 \left( \frac{R_2}{R_1 + R_2} \right) + V_H \left( \frac{R_1}{R_1 + R_2} \right) \quad \text{--- (ii)}$$

from Case 2:  $\Rightarrow 1.9 = 4 \left( \frac{R_2}{R_1 + R_2} \right) + V_L \left( \frac{R_1}{R_1 + R_2} \right) \quad \text{--- (iii)}$

[Note that  $V_x$  is not necessarily  $V_{TH}/V_{TL}$ ]

$$(ii) - (iii) = 3.8 = -2 \frac{R_2}{R_1 + R_2} + V_H \left( \frac{R_1}{R_1 + R_2} \right) - V_L \left( \frac{R_1}{R_1 + R_2} \right)$$

$$\Rightarrow 3.8 \Rightarrow -2 \frac{R_2}{R_1 + R_2} + 5 \Rightarrow \boxed{\frac{R_2}{R_1} = \frac{3}{2}} \quad \text{--- (iv)}$$

again  $R_x = \left( \frac{1}{R_1} + \frac{1}{R_2} \right)^{-1} = 2.9 \quad \text{--- (v)}$

solving (iv), (v),  $\boxed{R_1 = 4k\Omega} \quad \boxed{R_2 = 6k\Omega}$

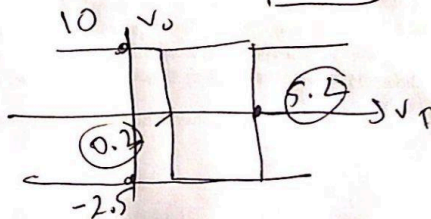
(b) Using  $R_1, R_2$  in (ii)  $\Rightarrow 5.2 = 2 \times \frac{6}{10} + V_H \frac{4}{10}$   
 $\Rightarrow \boxed{V_H = 10V}$

in (iii)  $\Rightarrow$  from (iii),  $V_L = \boxed{-2.5V}$

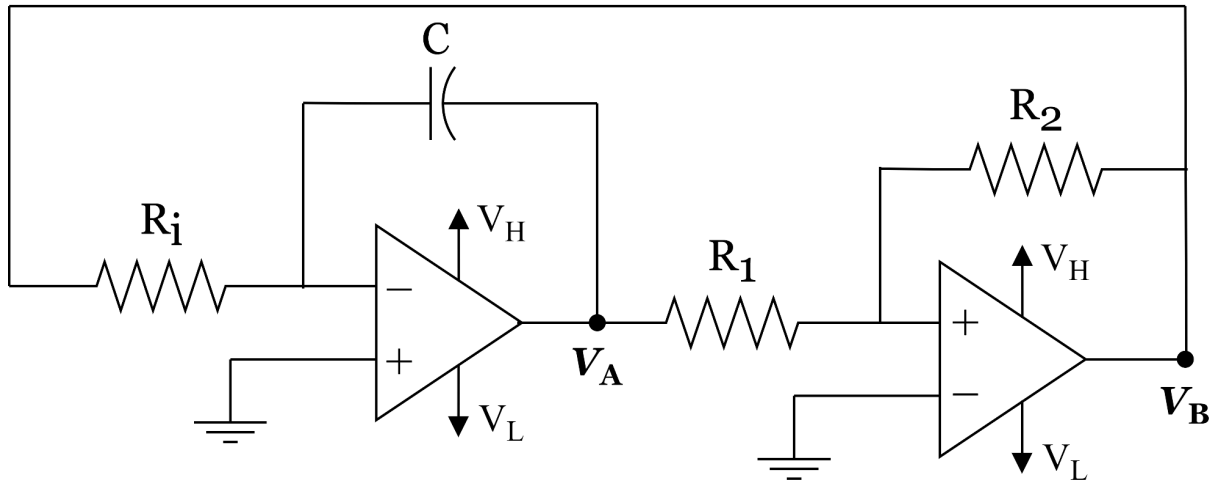
$\therefore$  These two voltages are possible.

(c) for switch position in A,

$$V_{TH} = \boxed{5.2V} \quad V_{TL} = 2 \times \frac{6}{10} + (-2.5) \times \frac{4}{10} = \boxed{0.2V}$$



## Signal Generator[QSH]:



For this Triangular Wave Generator, Given:

**[Set A]**  $V_H = 12\text{ V}$ ,  $V_L = -8\text{ V}$ ,  $R_1 = 6\text{ k}$ ,  $R_2 = 25\text{ k}$ ,  $C = 1\text{ }\mu\text{F}$ ,  $f = 5\text{ kHz}$

**[Set B]**  $V_H = 12\text{ V}$ ,  $V_L = -8\text{ V}$ ,  $R_1 = 12\text{ k}$ ,  $R_2 = 50\text{ k}$ ,  $C = 5\text{ }\mu\text{F}$ ,  $f = 1\text{ kHz}$

|     |     |                                                                                                                                                                                                                                                                  |             |
|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| (a) | CO1 | <b>Determine the duty cycle</b>                                                                                                                                                                                                                                  | <b>[4]</b>  |
| (b) | CO1 | <b>Draw the waveforms of <math>V_A</math> and <math>V_B</math> vs time in the same plot with appropriate labels (Values of <math>V_H</math>, <math>V_L</math>, <math>V_{TH}</math>, <math>V_{TL}</math>, <math>T_1</math>, <math>T_2</math>, <math>T</math>)</b> | <b>[6]</b>  |
| (c) | CO1 | <b>Calculate the value of <math>R_i</math></b>                                                                                                                                                                                                                   | <b>[10]</b> |

Solution

$$(a) \text{ Duty cycle} = \frac{T_1}{T_1 + T_2} = \frac{\frac{V_{TH} - V_{TL}}{V_H}}{\frac{V_{TH} - V_{TL}}{V_H} + \frac{V_{TL} - V_{TH}}{V_L}}$$

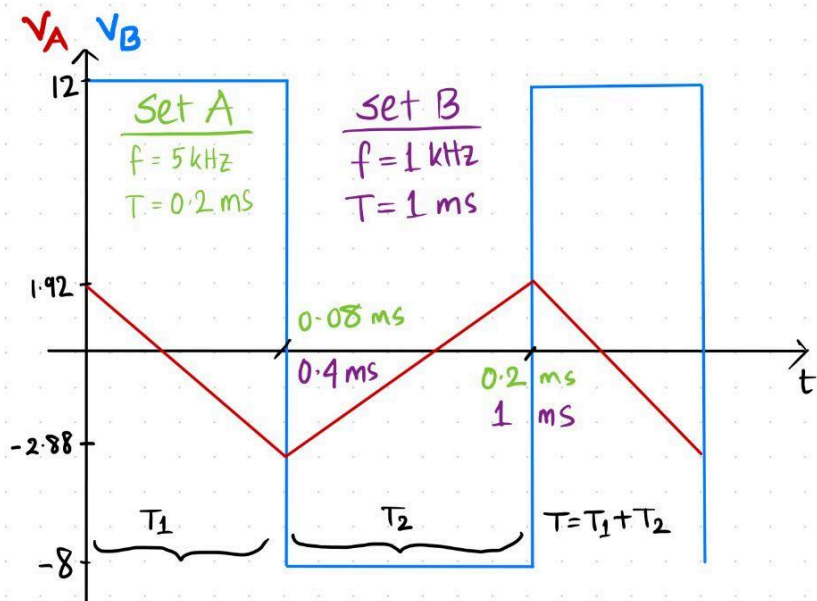
$$= 0.4 \rightarrow 40\%$$

$$(b) V_{TH} = -V_L \frac{R_1}{R_2}$$

$$= 1.92 \text{ V}$$

$$V_{TL} = -V_H \frac{R_1}{R_2}$$

$$= -2.88 \text{ V}$$



$$(c) T = T_1 + T_2$$

$$\Rightarrow \frac{1}{f} = R_i C \left( \frac{V_{TH} - V_{TL}}{V_H} + \frac{V_{TL} - V_{TH}}{V_L} \right)$$

$$\Rightarrow \frac{1}{f} = R_i C \times (V_{TH} - V_{TL}) \left( \frac{1}{V_H} - \frac{1}{V_L} \right)$$

$$\Rightarrow R_i = \left[ f C \frac{R_1}{R_2} (V_H - V_L) \left( \frac{1}{V_H} - \frac{1}{V_L} \right) \right]^{-1}$$

$$= \left[ 5 \times 10^{-3} \times \frac{6}{25} \times (12 + 8) \left( \frac{1}{12} + \frac{1}{8} \right) \right]^{-1}$$

$$= 200 \Omega$$

# CMOS and MOSFET- Jishnu Mahmud

## Question 4

The NMOS NOR gate in **figure 5** is biased at  $V_{DD} = 5V$ . The transistor parameters are  $K_n = 0.3 \text{ mA/V}^2$ ,  $V_{TN1} = 0.5V$ ,  $V_{TN2} = 0.8V$ .  
For triode mode,  $i_D = k_n [2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2]$ , and for saturation mode,  $i_D = k_n [V_{GS} - V_{TN}]^2$   
[20 + 2bonus]

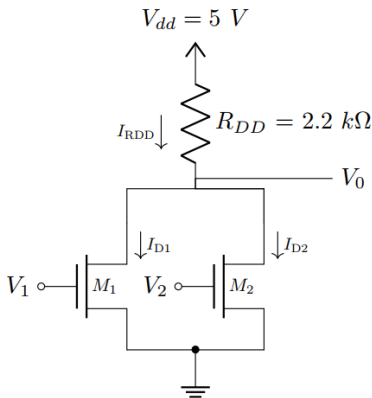


Figure 5: Circuit

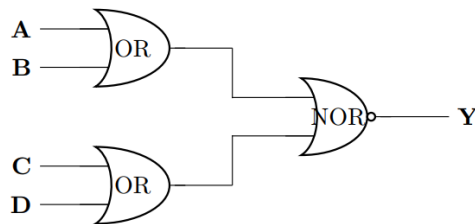


Figure 6

|     |     |                                                                                                                                   |     |
|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------|-----|
| CO1 | (a) | <b>Find</b> the voltage, $V_0$ , for all input combinations and make a truth table for the circuit in <b>figure 5</b> .           | [6] |
| CO1 | (b) | <b>Identify</b> the boolean function, $Y$ , of the logic circuit in <b>figure 6</b> .                                             | [2] |
| CO1 | (c) | Hence, <b>design</b> the CMOS circuit for $Y$ . (Clearly show the PMOS and NMOS networks).                                        | [8] |
| CO1 | (d) | Now, <b>design</b> the boolean function, $Y$ , from <b>part (b)</b> using multiple copies of the circuit in <b>figure 5</b> only. | [4] |

Use the CMOS circuit of figure 4 to design the boolean function of  $Y$ , from part (b) [Hint: Use the figure 4 multiple times to design  $Y$ ]

set A

| $V_1$ | $V_2$ | $V_0$  |
|-------|-------|--------|
| 0     | 0     | 5      |
| 0     | 5     | 0.0895 |
| 5     | 0     | 0.0835 |
| 5     | 5     | 0.0433 |

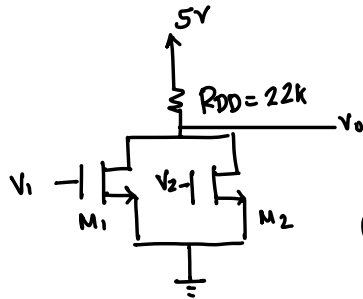
set B

| $V_1$ | $V_2$ | $V_0$  |
|-------|-------|--------|
| 0     | 0     | 6      |
| 0     | 6     | 0.096  |
| 6     | 0     | 0.09   |
| 6     | 6     | 0.0467 |

SETA  
Question 4

Spring '24  
CSE 350  
BRACU  
Jishnu Mahmud

(a) NOR



(i)  $V_1, V_2 = 0$

$V_o = 5V$ ,  $M_1, M_2$  OFF

(ii)  $V_1, V_2 = 5$   $V_o = V_D = \text{small}$   $V_G = \text{large}$

$V_{DS} < V_{GS} - V_T$

Triode

$$\begin{aligned} V_{TN1} &= 0.5V \\ V_{TN2} &= 0.8V \\ k_n &= 0.3 \text{ mA/V}^2 \end{aligned}$$

$$\begin{aligned} &0.3 \left[ 2(5 - 0.5)V_o - V_o^2 \right] + 0.3 \left[ 2(5 - 0.8)V_o - V_o^2 \right] \\ &= 0.3 \left[ 10V_o - V_o - V_o^2 \right] + 0.3 \left[ 10V_o - 1.6V_o - V_o^2 \right] \\ &= 0.3 \left\{ 9V_o - V_o^2 + 8.4V_o - V_o^2 \right\} \\ &= 0.3 \left\{ -2V_o^2 + 17.4V_o \right\} \end{aligned}$$

$$\begin{aligned} 0.3 \left\{ -2V_o^2 + 17.4V_o \right\} &= \frac{5 - V_o}{22} \\ (-0.6V_o^2 + 5.22V_o)22 - 5 + V_o &= 0 \\ -13.2V_o^2 + 115.84V_o - 5 &= 0 \end{aligned}$$

$$V_o = 8.73V \quad V_o = 0.0433V$$

$\times$   $\checkmark$

$$V_{GS2} - V_{T2} = 5 - 0.8 > 0.043$$

$$V_{GS1} - V_{T1} = 5 - 0.5 > 0.043$$

$\therefore$  Triode  $\checkmark$

$$\begin{aligned} \textcircled{iii} \quad 0.3 \left[ 2(5 - 0.5)V_o - V_o^2 \right] &= \frac{5 - V_o}{22} \\ 22 \left( 2.7V_o - 0.3V_o^2 \right) &= 5 - V_o \\ -6.6V_o^2 + 59.4V_o + V_o - 5 &= 0 \end{aligned}$$

$$-6.6V_o^2 + 60.4V_o - 5 = 0$$

$$V_o = 9.1 \quad V_o = 0.0835V$$

$\times$   $\checkmark$

$$V_{GS1} - V_{T1} = 5 - 0.8 > 0.0835$$

Triode

$$\textcircled{iv} \quad 0.3 \left[ 2(5 - 0.8)V_o - V_o \right] = \frac{5 - V_o}{22}$$

$$-6.6V_o + 55.44V_o + V_o - 5 = 0$$

$$-6.6V_o + 56.44V_o - 5 = 0$$

$$V_o = 8.46V \quad V_o = 0.0895V$$

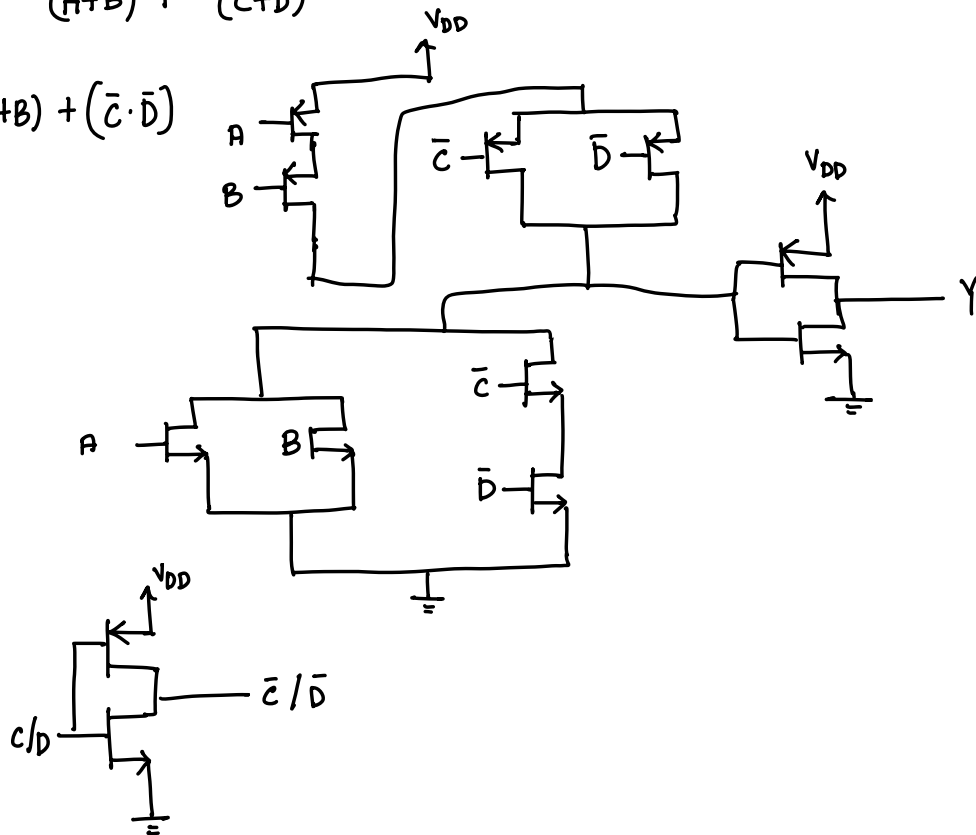
x                      ✓

$$V_{GS} - V_T > 0.0895$$

NMOS

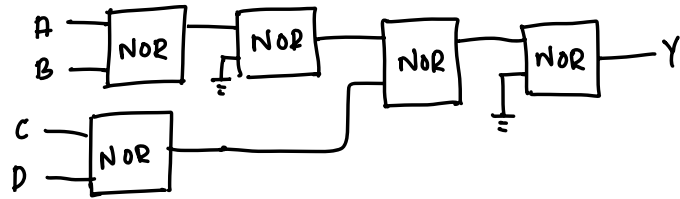
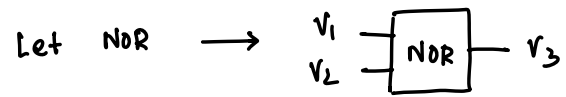
$$(b) \quad Y = (A+B) + (\overline{C+D})$$

$$(c) \quad (A+B) + (\overline{C} \cdot \overline{D})$$

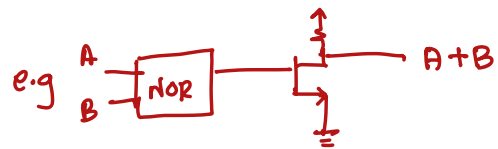


\* please allow full marks to any intelligent attempt to this question even if final answer is wrong.

(d)



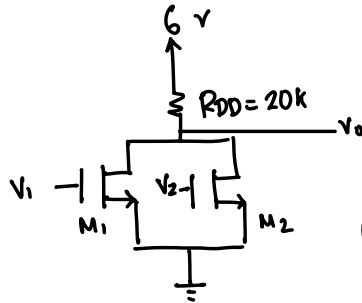
\* allow full marks even if student uses inverter after NOR



SETB  
Question: 03

changes:  $\left\{ V_{DD} = 6V, R_{DD} = 20k\Omega \right\}$

(a) NOR



(i)  $V_1, V_2 = 0$

$V_o = 6V$ ,  $M_1, M_2$  OFF

(ii)  $V_1, V_2 = 6$   $V_o = V_D = \text{small}$   $V_G = \text{large}$

$$V_{DS} < V_{GS} - V_T$$

Triode

$$\begin{aligned} V_{TN1} &= 0.5V \\ V_{TN2} &= 0.8V \\ k_n &= 0.3 \text{ mA/V}^2 \end{aligned}$$

$$\begin{aligned} &0.3 \left[ 2(6 - 0.5)V_o - V_o^2 \right] + 0.3 \left[ 2(6 - 0.8)V_o - V_o^2 \right] \\ &= 0.3 \left[ 12V_o - V_o - V_o^2 \right] + 0.3 \left[ 12V_o - 1.6V_o - V_o^2 \right] \\ &= 0.3 \left\{ 11V_o - V_o^2 + 10.4V_o - V_o^2 \right\} \\ &= 0.3 \left\{ -2V_o^2 + 21.4V_o \right\} \end{aligned}$$

$$0.3 \left\{ -2V_o^2 + 21.4V_o \right\} = \frac{6 - V_o}{20}$$

$$-12V_o^2 + 128V_o + V_o - 6 = 0$$

$$V_o = 10.7V \quad V_o = 0.0467$$

$$V_{GS2} - V_{T2} = 5 - 0.8 > 0.0467V$$

$$V_{GS1} - V_{T1} = 5 - 0.5 > 0.0467V$$

$\therefore$  Triode ✓

$$(ii) \quad 0.3 \left[ 2(6 - 0.5)V_o - V_o^2 \right] = \frac{6 - V_o}{20}$$

$$66V_o + V_o - 6V_o^2 - 6 = 0$$

$$V_o = 11.1 \quad V_o = 0.090V$$

$$V_{GS1} - V_{T1} = 5 - 0.8 > 0.090V$$

Triode ✓

$$\textcircled{iv} \quad 0.3 \left[ 2(6 - 0.8)V_o - V_o^2 \right] = \frac{6 - V_o}{20}$$

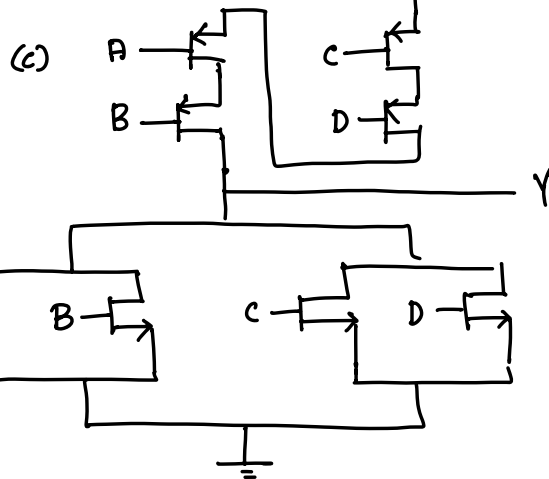
$$62.4V_o - 6V_o^2 + V_o - 6 = 0$$

$$V_o = 10.5V, \quad V_o = 0.096V$$

$$V_{as} - V_T > 0.096V$$

Niodc

(b)  $(A+B) + (C+D)$



(d)

